Sede Nube (AWS 2):Análisis de seguridad con Lynis

Objetivo

Antes de aplicar cualquier medida de hardening, se ejecuta Lynis con el fin de obtener una visión objetiva del estado de seguridad del sistema.
El objetivo es establecer una línea base medible que permita comparar posteriormente el impacto real de las mejoras aplicadas, evitando suposiciones sin datos.

Información del sistema

- Sistema operativo: Ubuntu 24.04.3 LTS
- Contexto: Servidor en funcionamiento con servicios activos en producción

```
Last login: Sun Mar 29 15:31:09 2026 from 18.206.107.27
ubuntu@ip-10-1-0-85:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.3 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.3 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
```

Estado inicial del sistema

Servicios detectados:

- Docker y containerd en ejecución
- Apache desplegado dentro de contenedor
- No se detecta fail2ban instalado
- No se detecta rkhunter instalado

```
ubuntu@ip-10-1-0-85:~$ systemctl list-units --type=service --state=running
  UNIT                                          LOAD   ACTIVE SUB     DESCRIPTION
  acpid.service                                 loaded active running ACPI event daemon
  chrony.service                                loaded active running chrony, an NTP client/server
  containerd.service                            loaded active running containerd container runtime
  cron.service                                  loaded active running Regular background program processing daemon
  dbus.service                                  loaded active running D-Bus System Message Bus
  docker.service                                loaded active running Docker Application Container Engine
  getty@tty1.service                            loaded active running Getty on tty1
  irqbalance.service                            loaded active running irqbalance daemon
  ModemManager.service                          loaded active running Modem Manager
  multipathd.service                            loaded active running Device-Mapper Multipath Device Controller
  networkd-dispatcher.service                   loaded active running Dispatcher daemon for systemd-networkd
  polkit.service                                loaded active running Authorization Manager
  rsyslog.service                               loaded active running System Logging Service
  serial-getty@ttyS0.service                    loaded active running Serial Getty on ttyS0
  snap.amazon-ssm-agent.amazon-ssm-agent.service loaded active running Service for snap application amazon-ssm-agent.amazon-ssm-agent
  snapd.service                                 loaded active running Snap Daemon
  ssh.service                                   loaded active running OpenBSD Secure Shell server
  systemd-journald.service                      loaded active running Journal Service
  systemd-logind.service                        loaded active running User Login Management
  systemd-networkd.service                      loaded active running Network Configuration
  systemd-resolved.service                      loaded active running Network Name Resolution
  systemd-udevd.service                         loaded active running Rule-based Manager for Device Events and Files
  udisks2.service                               loaded active running Disk Manager
  unattended-upgrades.service                   loaded active running Unattended Upgrades Shutdown
  user@1000.service                             loaded active running User Manager for UID 1000

Legend: LOAD   → Reflects whether the unit definition was properly loaded.
        ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
        SUB    → The low-level unit activation state, values depend on unit type.

25 loaded units listed.
ubuntu@ip-10-1-0-85:~$
```

Puertos abiertos:

- Puerto 22 (SSH) accesible desde 0.0.0.0 (expuesto a cualquier origen)
- Puertos 80 y 443 abiertos para servicio web (Apache)

Observación:

- Se identifica como mejora prioritaria la restricción de acceso al servicio SSH

```
ubuntu@ip-10-1-0-85:~$ ss -tlnp
State                  Recv-Q                 Send-Q                                 Local Address:Port
LISTEN                 0                      4096                                       127.0.0.1:40493
LISTEN                 0                      4096                                    127.0.0.54:53
LISTEN                 0                      4096                                 127.0.0.53%lo:53
LISTEN                 0                      4096                                       0.0.0.0:22
LISTEN                 0                      4096                                       0.0.0.0:80
LISTEN                 0                      4096                                          [::]:22
LISTEN                 0                      4096                                          [::]:80
```

Resultados del análisis Lynis

Malware scanner : NO instalado [✗]
Firewall : Activo [✓]
Hardening index : 60 / 100

```
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Lynis security scan details:

Hardening index : 60 [############        ]
Tests performed : 258
Plugins enabled : 1

Components:
- Firewall               [V]
- Malware scanner        [X]

Scan mode:
Normal [V]  Forensics [ ]  Integration [ ]  Pentest [ ]

Lynis modules:
- Compliance status      [?]
- Security audit         [V]
- Vulnerability scan     [V]

Files:
- Test and debug information      : /var/log/lynis.log
- Report data                     : /var/log/lynis-report.dat

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Lynis 3.0.9

Auditing, system hardening, and compliance for UNIX-based systems
(Linux, macOS, BSD, and others)

2007-2021, CISOfy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)

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[TIP]: Enhance Lynis audits by adding your settings to custom.prf (see /etc/lynis/default.prf for all settings)
```